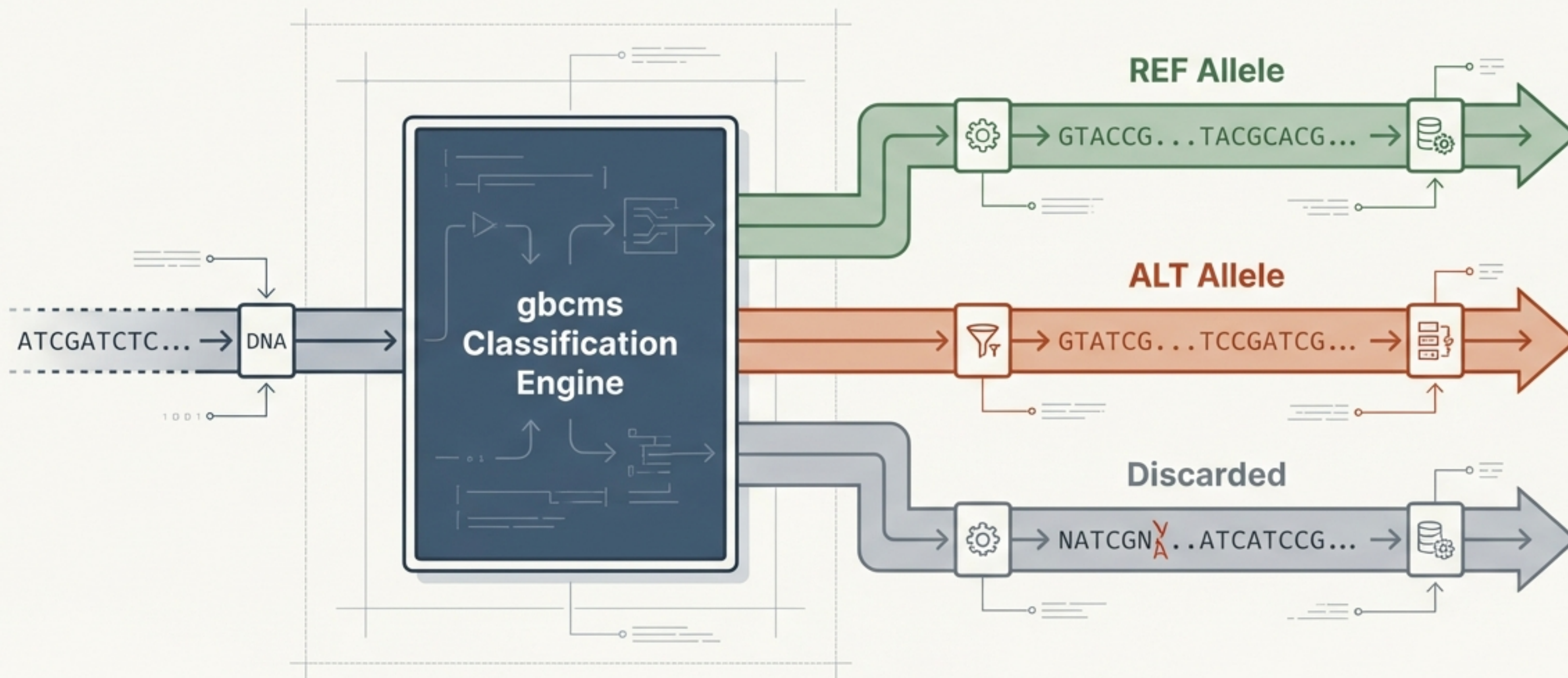


Inside the Engine: Allele Classification Phases

A structural deep-dive into the algorithmic cascade and internal heuristics of the **gbcms** variant counting engine.



The Variant Counting Compromise

Speed



ATCGGT...CTAG
ATCGGT...CTAG

Naive Exact Matching

Computationally cheap, highly scalable. Fails catastrophically in complex, repetitive, or low-quality genomic regions.

Phase 1 & 2:
Exact & Masked Fast-Paths

Phase 2.5:
Edit Distance Rescue

Phase 3:
SW Alignment Fallback

The Solution: A multi-tiered algorithmic cascade. The engine routes progressively heavier phases, reserving full dynamic programming only for the edge cases that strictly require it.

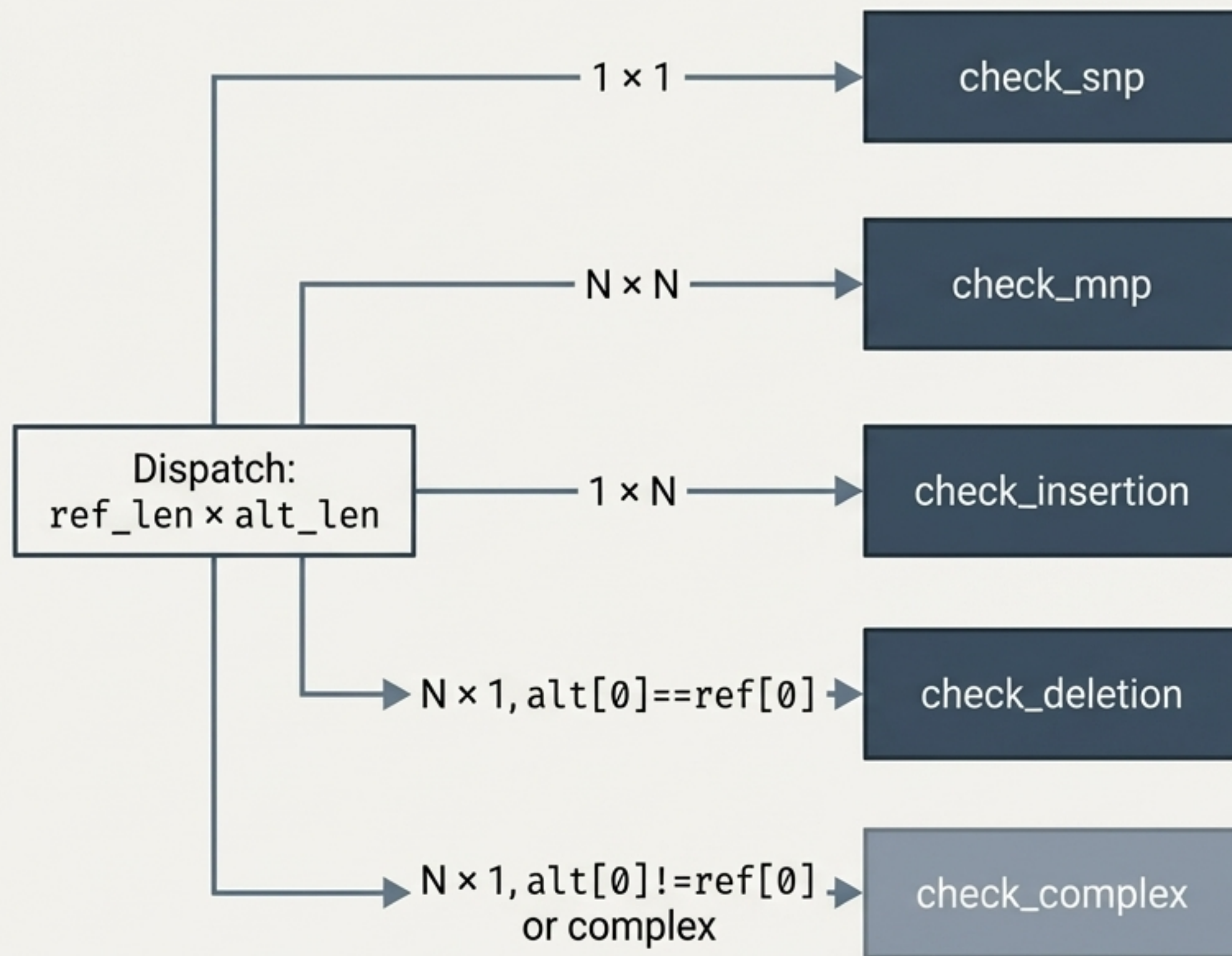
Accuracy

	A		C		T		G
A	1	→	↓		0		
T	↖		3	→	↓	-1	1
		0	↖		3	→	↓
C	1			0	↖	4	5
G					1	↖	→

Smith-Waterman PairHMM

Perfect structural resolution. Computationally impossible to run on every read across whole genomes.

The Dispatch Switchboard



The Nx1 Edge Case (Anchor Substitution)

$\text{alt_allele}[0] \neq \text{ref_allele}[0]$

Dispatch relies on allele lengths. For $N \times 1$ variants, it checks if the anchor base substitutes. If substituted (e.g., GC $\rightarrow$ T), it routes to `check_to` to prevent left-alignment CIGAR safeguards from misclassifying simultaneous anchor substitutions.

The Algorithmic Cascade



Phase 1

Normalizes the raw read alignment into a genomic footprint.

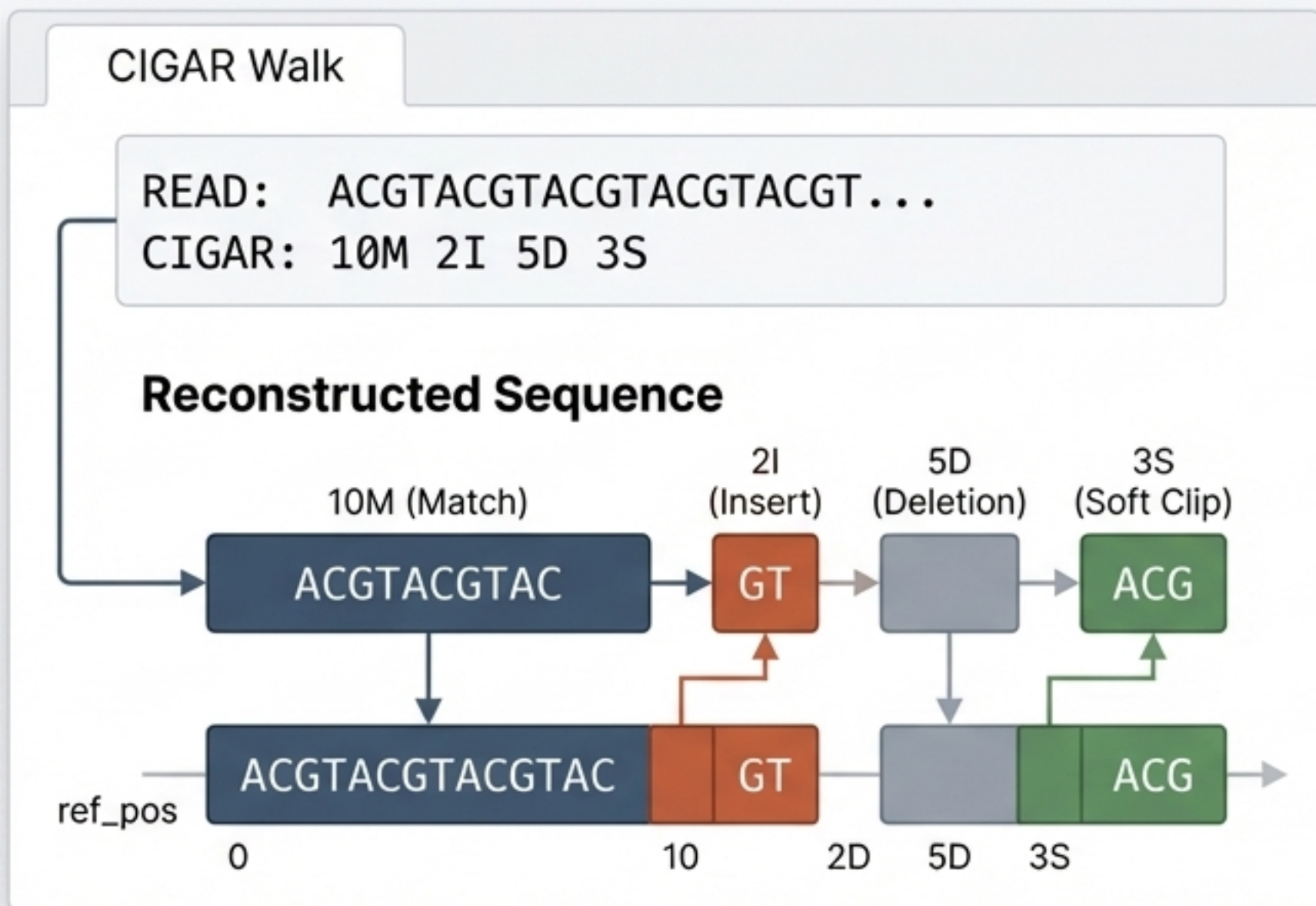
Phase 2 & 2.5

The primary fast-path classification engine. Successfully resolves >90% of reads.

Phase 3

The heavy-lifting rescue path. Triggers only for structural mismatches, slippery repeats, and severe noise.

Phase 1: Haplotype Reconstruction

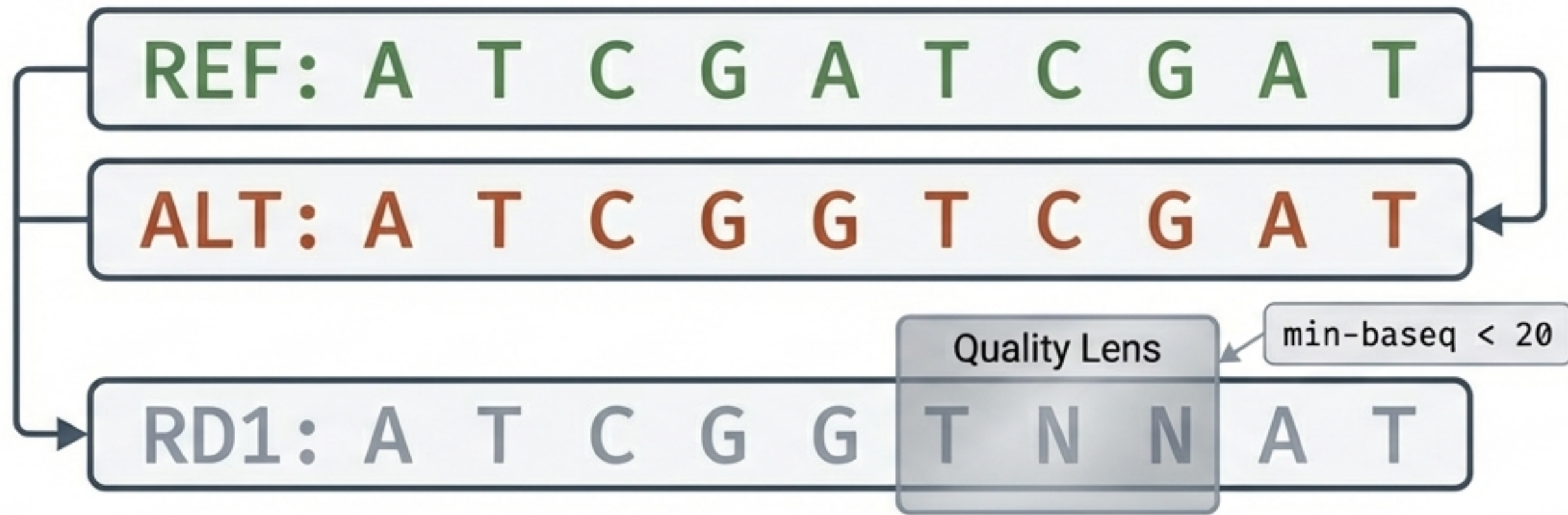


CIGAR Operation Translation	
CIGAR Op	Action
M / = / X	Append overlapping bases and qualities
I	Append inserted bases if ref_pos is within variant region
D / N	Advance ref_pos without appending
S	Append if ref_pos overlaps variant window
H / P	No action



Crucial Recovery: `Soft-clipped (S)` evidence is actively recovered and appended if it overlaps the genomic window, rescuing reads that strict aligners discard.

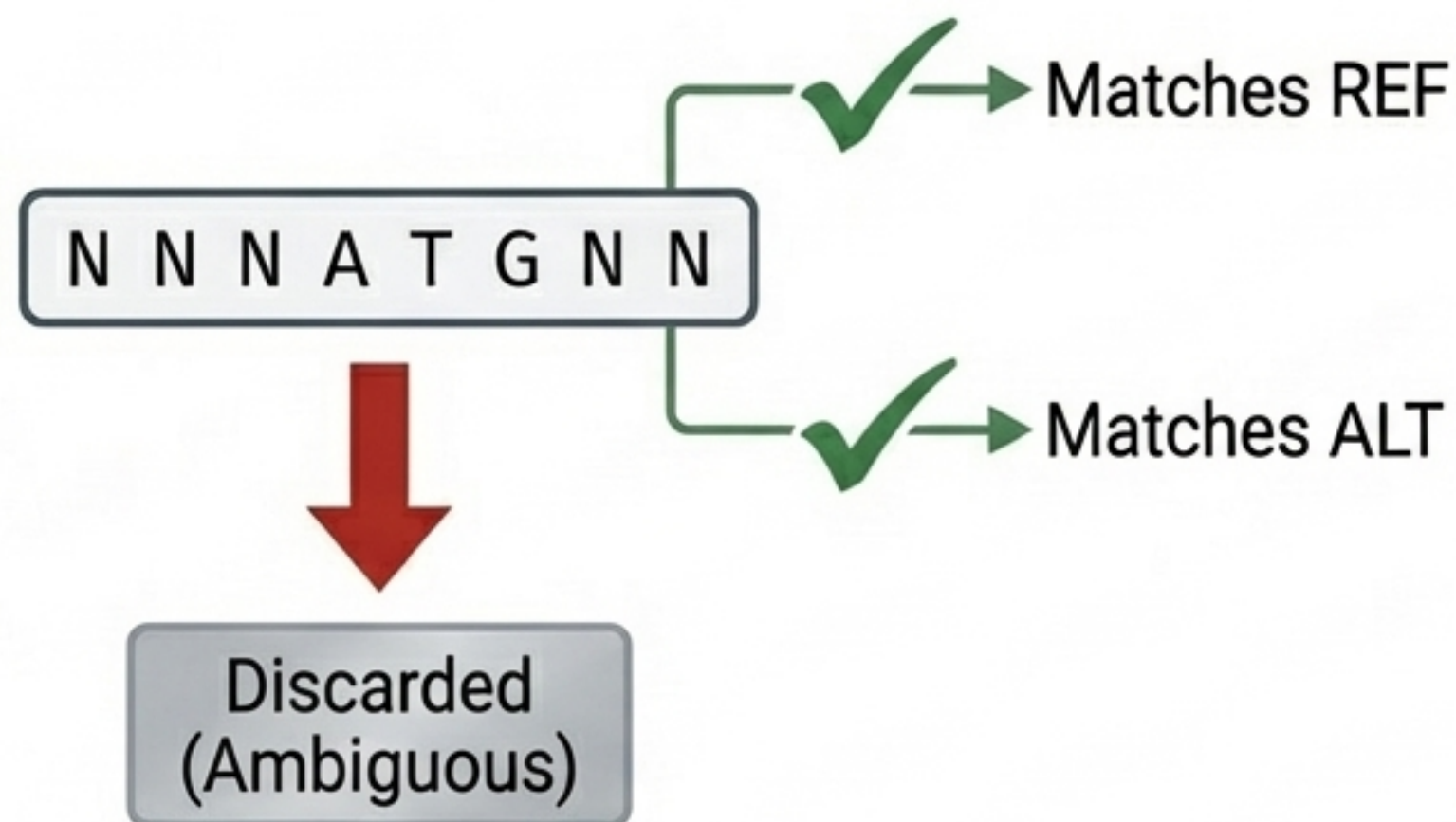
Phase 2: Masked Comparison



- Instead of exact matching, bases with quality below --min-baseq are masked out. They are treated as wildcards.
- Low-quality sequencing noise is stripped of its identity, ensuring it cannot 'vote' for either allele, preventing false positives in complex structural variants.

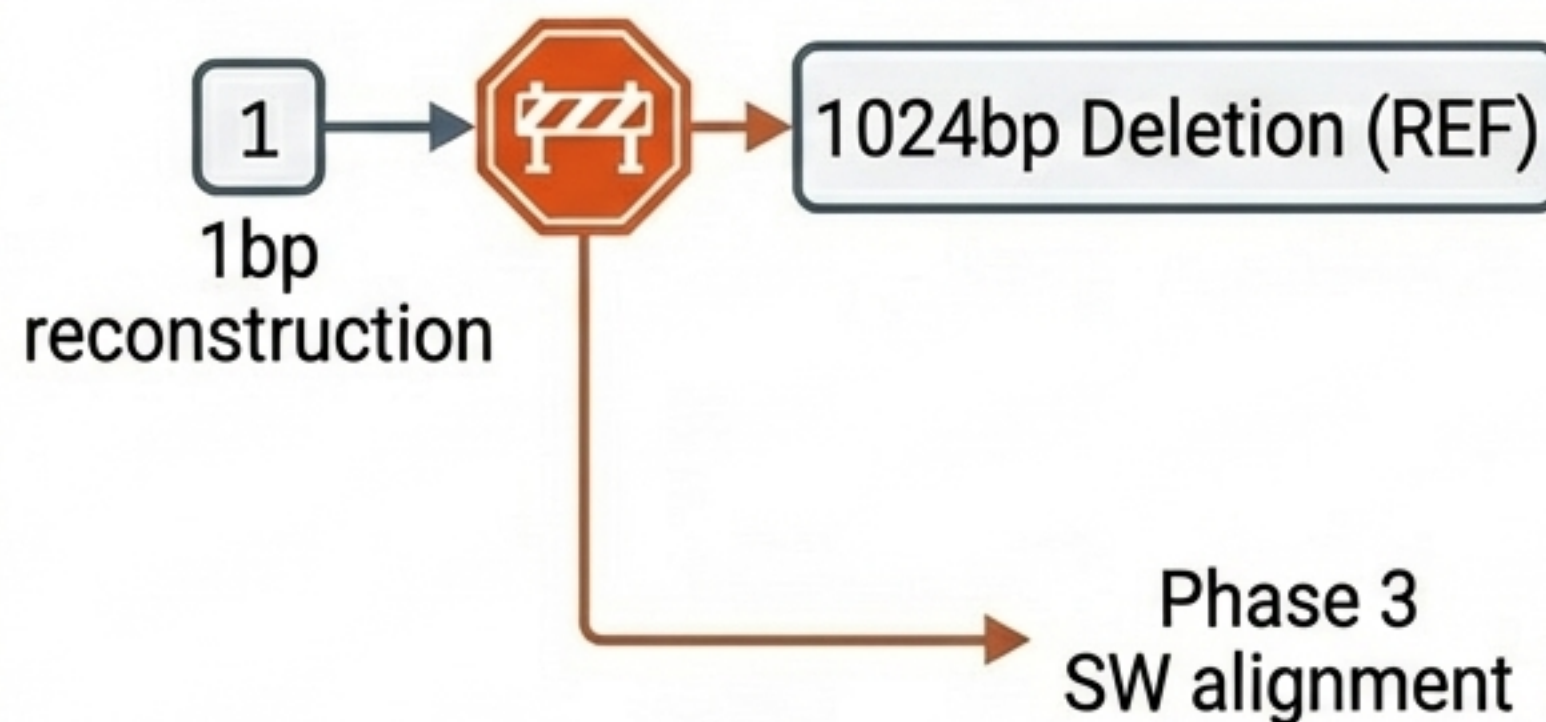
Phase 2: Ambiguity & Guardrails

Ambiguity Detection



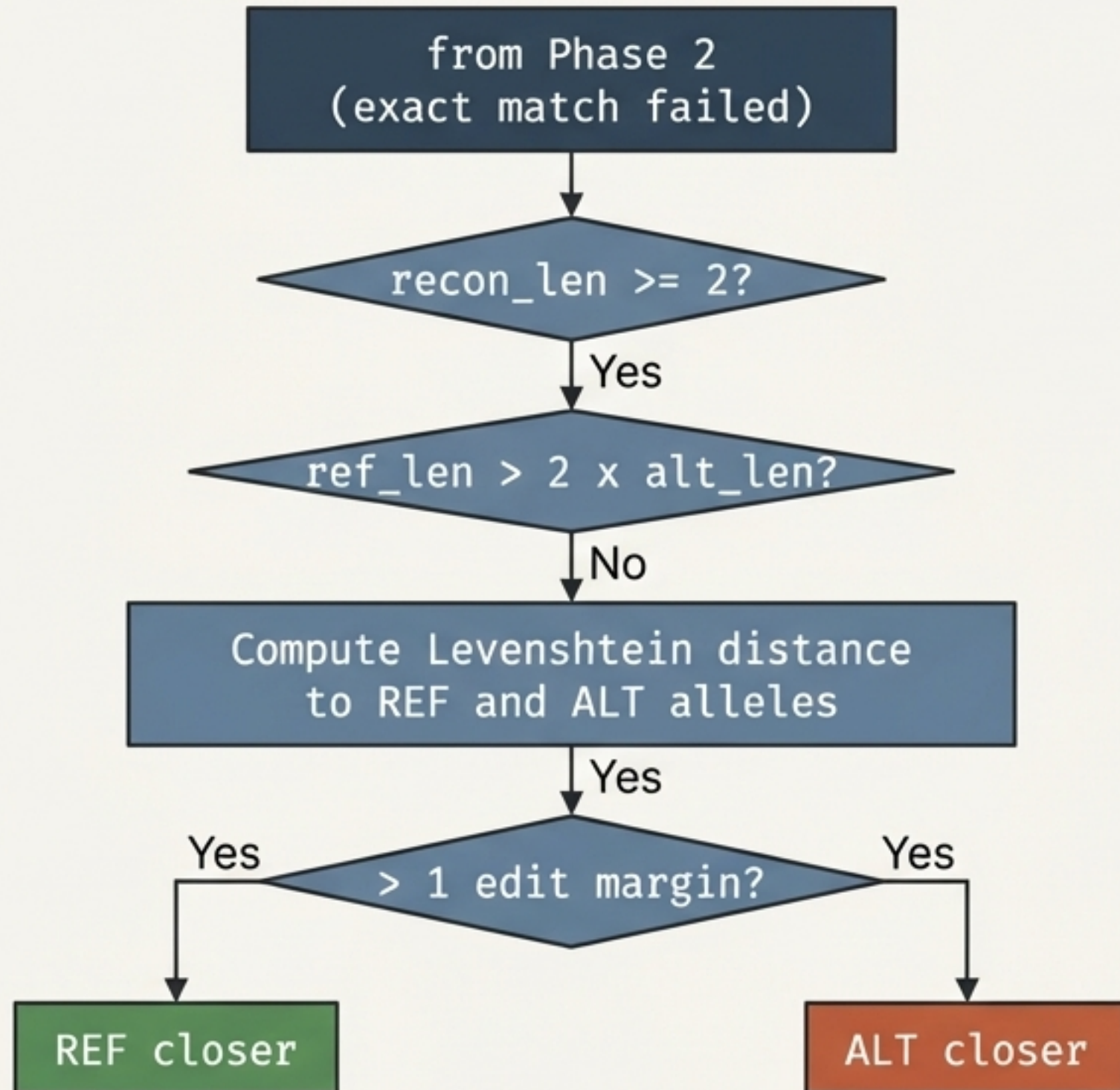
If masking causes reliable bases to match BOTH alleles, the read is discarded rather than guessed to prevent false calls.

Large-REF Guard



If a reconstruction is <10% of a massive REF length, it is blocked from a trivial match and routed directly to Phase 3 SW alignment.

Phase 2.5: The Edit Distance Fast-Path



Trigger:	Phase 2 strict length comparison fails.
Mechanism:	Measures Levenshtein edit distance between the reconstruction and each allele.
Guardrail:	Requires >1 edit margin between REF and ALT distances to confidently assign an allele.
Result:	Rescues reads with minor sequencing noise (1-2 base errors) on incomplete variant incomplete variant definitions without incurring massive PairHMM computational penalties.

Triggering Phase 3: The Alignment Fallback

Diagnostic Dashboard

Structural: CIGAR Disconnect

Read position cannot be found during the CIGAR walk.

Structural: Incomplete Span

Read does not cover the entire MNP region.

Contiguity: Internal Indel

An indel is detected directly inside an MNP block.

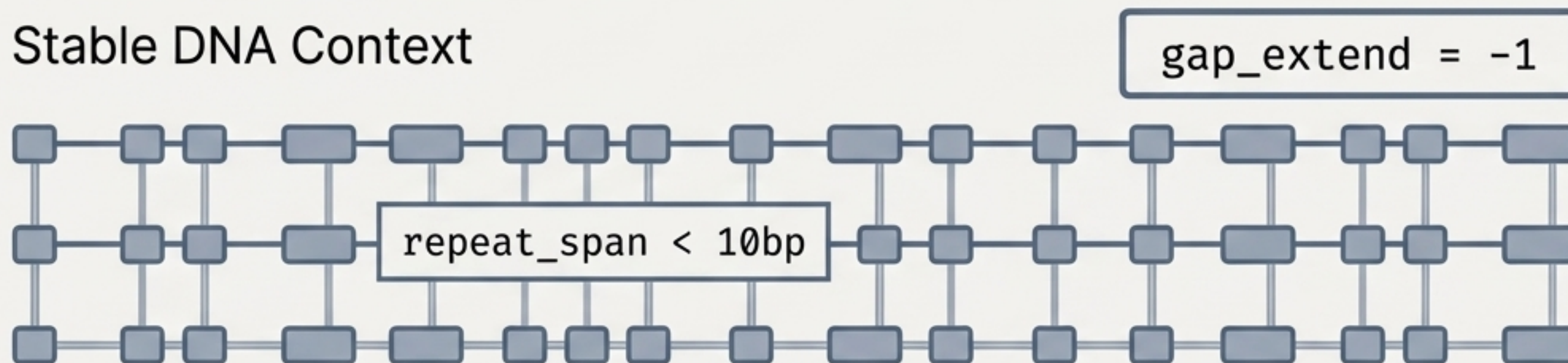
Quality: Catastrophic Noise

$\min(\text{BQ across block})$ drops below the required threshold.

EJECTED FROM FAST-PATH. Sent directly to WFA → PairHMM Aligners for rigorous, dual-haplotype dynamic programming resolution.

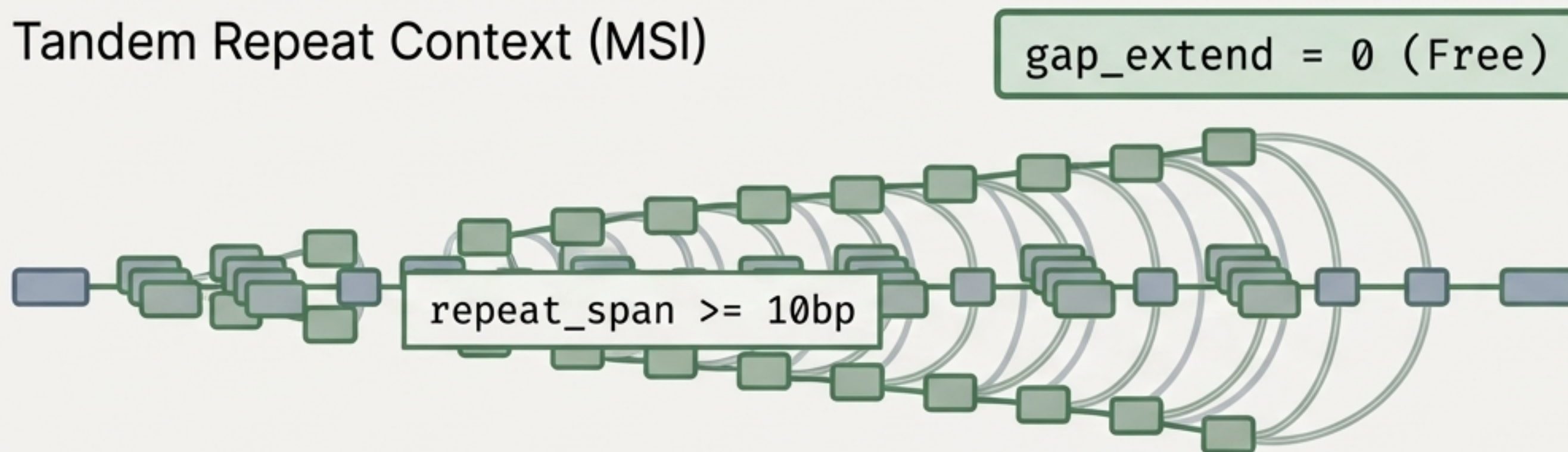
Phase 3: Dynamic SW Gap Penalties

Stable DNA Context



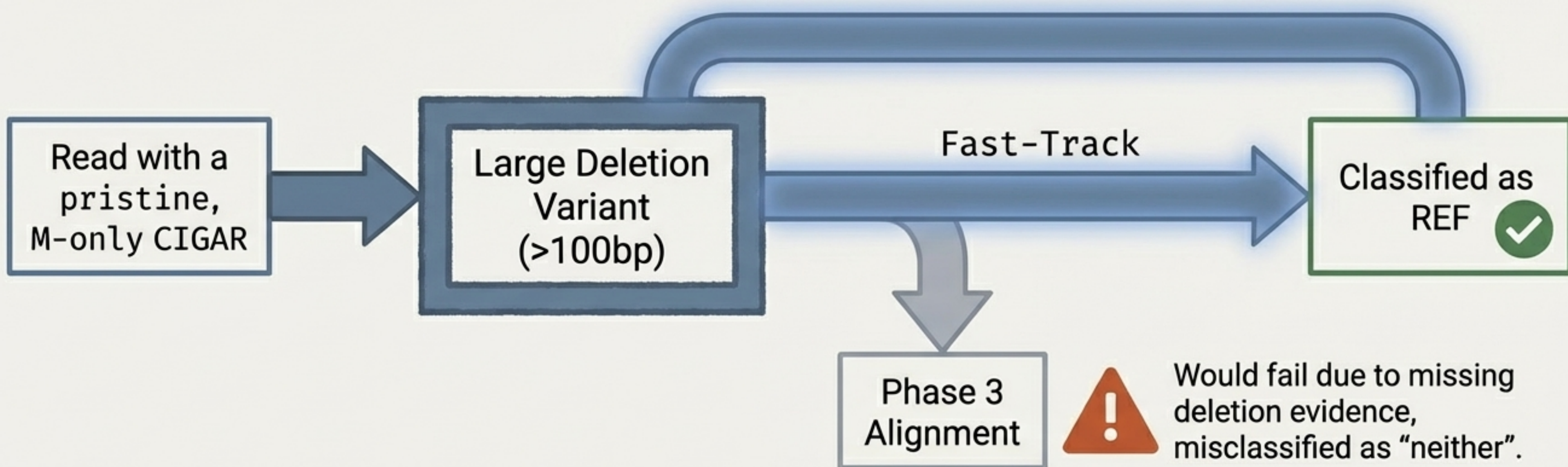
Standard penalty.
Prevents spurious and artificial gap extensions in normal genomic regions.

Tandem Repeat Context (MSI)



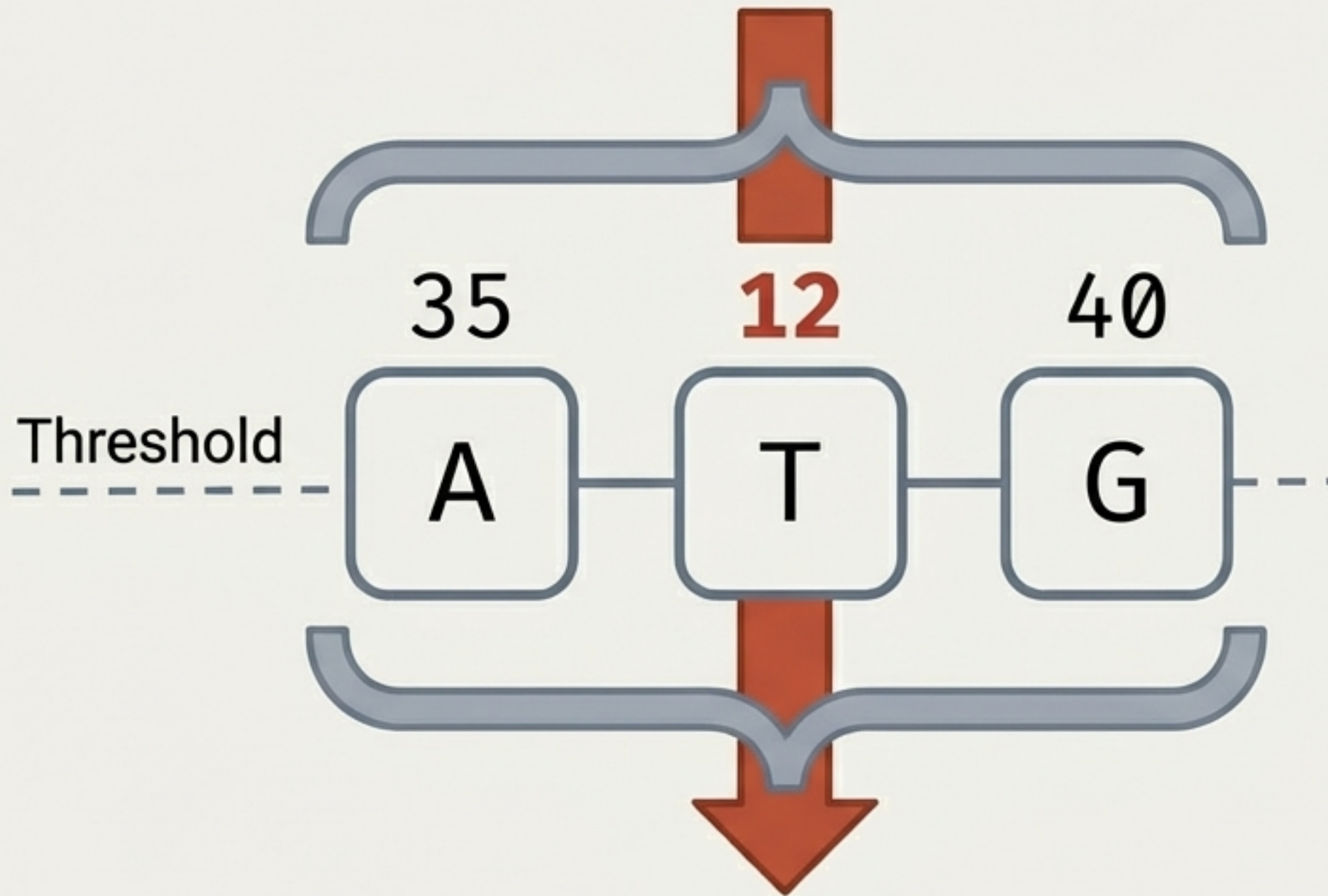
Adaptive penalty.
Natively absorbs polymerase slippage noise by making gap extensions free, preventing severe undercounting in repetitive MSI regions.

The REF Fallback for Large Deletions



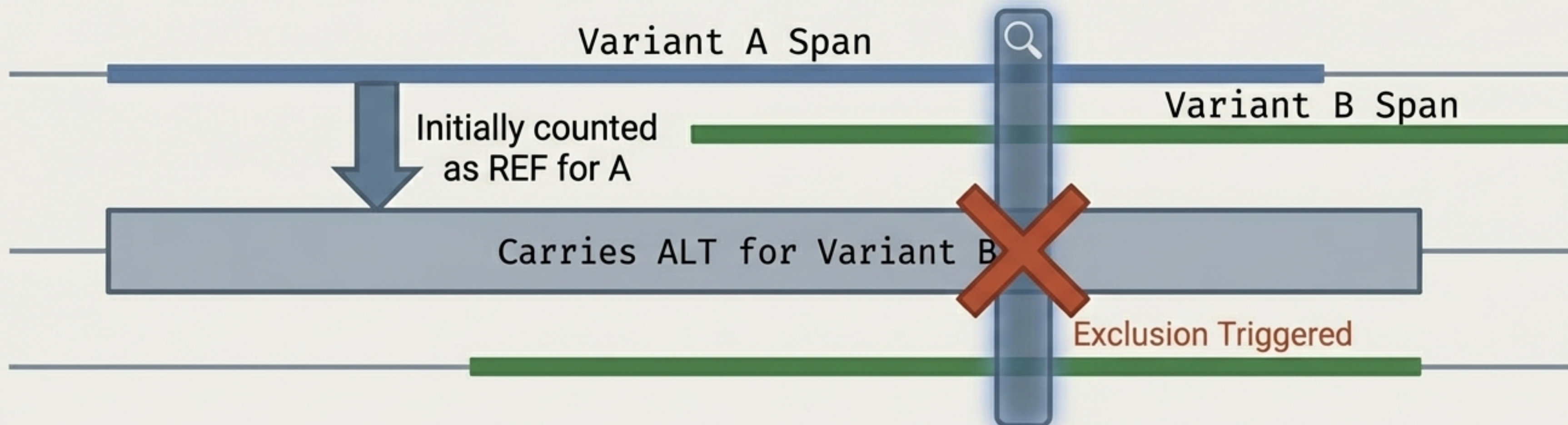
- **The Threat:** Clean REF reads naturally lack CIGAR indel evidence for massive structural deletions. If subjected to Phase 3 Smith-Waterman, they are heavily penalized and dropped.
- **The Rescue:** If the engine detects any clean M-only block covering the anchor position of a large deletion, Phase 3 is bypassed and the read is confidently saved as REF.

MNP Strategy: Min-BQ-Across-Block



- **Contiguous Quality Enforcement:** MNP quality is assessed using the absolute minimum base quality across the entire block across Fira Code.
- If **ANY** base in an MNP falls below the threshold, the exact match is skipped entirely for the whole contiguous sequence.
- This prevents catastrophic ALT loss, routing borderline reads directly to Phase 3 where quality-masking can safely rescue reliable bases.

Multi-Allelic Behavior & Sibling Exclusion



The Reality of Complex Loci

Reads carrying an ALT allele for one variant can easily be miscounted as REF for a nearby overlapping variant in multi-allelic regions.

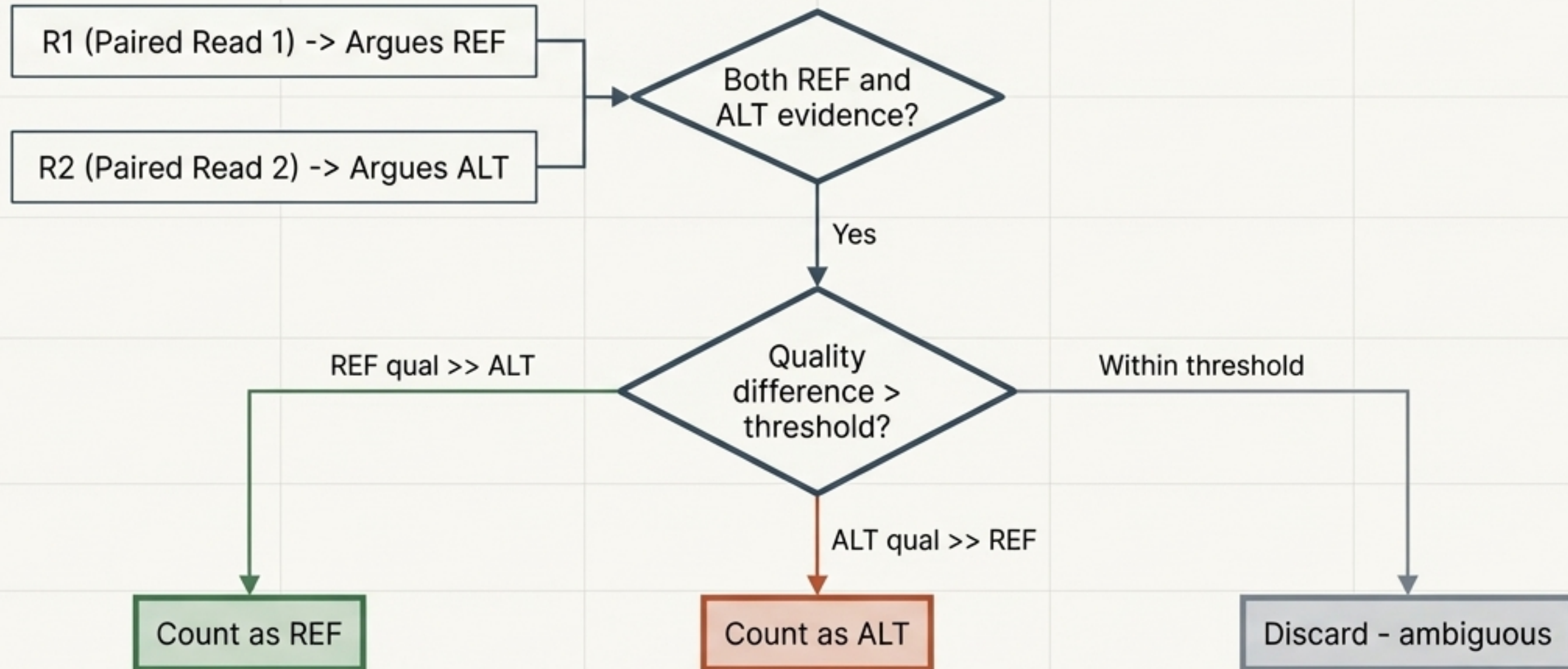
Sibling ALT Exclusion

Any read classified as REF is actively checked against all sibling variants in the same spatial group. If it carries a sibling's ALT allele, it is excluded from the REF count, preserving unbiased Variant Allele Frequency (VAF).

The Phase Resolution Matrix

Phase	Trigger Condition	Algorithmic Approach	Computational Cost	Primary Target
Phase 1	Raw alignment	CIGAR Walk	$O(1)$	Normalizing genomic footprint
Phase 2	Reconstructed string	Masked sequence match	Low	>90% of high-quality SNPs/Indels
Phase 2.5	Length match failure	Levenshtein Edit Distance	Medium	Incomplete variant definitions
Phase 3	Structural/Quality failure	WFA & PairHMM	High	Complex Del+SNVs, Slippery repeats

The Final Output: Fragment Quality Consensus



The classification journey ends at the fragment level. When paired-end reads disagree, the engine executes a quality-weighted consensus. It assigns the entire fragment to the allele supported by the highest base quality margin, resulting in pristine Depth (DP) and Allele Depth (AD) metrics.